Gravity in AI-to-AI loops (working draft)

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Brad Hinkel, with Claude (Anthropic). First draft written by Claude Fable 5.1 on 2026-10-23 from the gate reports in the ai_to_ai_loops repository; review and revision by Brad Hinkel in progress. Brad takes responsibility for the claims.

Summary

Attractors in caption-to-image-to-caption loops are temporary stops, not endpoints. Every basin we measured has a finite escape rate, trajectories leave and come back, and the loop as a whole is a stationary process that tours a small family of states. Hintze and colleagues showed in 2025 that these loops converge to generic visual motifs and described the motifs as robust basins of attraction. This draft reports what happens when you measure the basins: how often a loop falls in (radius), how long it stays (depth), what decides which basins exist, and what we tried in order to keep a loop out.

This is a working draft by an independent AI experimenter, not a submission. I reused Hintze's open code and seed prompts, ran the loops on rented GPUs with Claude Code as the engineer and analyst, and pre-registered a prediction before every run. About 90,000 loop steps across twelve runs sit behind the numbers here. I am circulating it to start a conversation with people who study these loops, and I would welcome corrections, reanalysis and co-authors.

Four findings:

1. **Temporary stops.** The strongest basin we found, a gothic cathedral nave, holds a trajectory for 30 steps on average and still loses it at 3.3% per step. The most popular basin, a palatial salon, holds for 9 steps at 11% per step. Escape rates do not fall as the run goes on, so nothing is settling.
2. **Radius and depth are measurable, and depth is a profile.** Capture rate and endpoint share give the radius. Escape hazard and residence give the depth. Inside a basin the exit rate rises tenfold from the centre to the rim.
3. **The landscape belongs to the model pair.** The image generator sets the image-space attractors and the captioner sets the caption-space attractors. The same motif appears in several pairs, but its depth does not carry over. The caption instruction and its language reshape the landscape by amplifying modes that were already latent.

4. **Nothing we tried keeps a loop out.** Rejecting images near already-visited states cut stays by two to three times, and the trajectories toured the same family anyway. An instruction added for measurement created a new attractor of its own. The paper closes with what to try next.

Background: what Hintze asserted and what was left to measure

An AI-to-AI loop takes a text prompt, renders it with an image generator, has a vision-language model describe the image, and feeds the description back as the next prompt. Hintze, Proschinger Åström and Schossau ran this with Stable Diffusion XL and LLaVA from 100 seed prompts and found that trajectories lose their seed within a few steps and settle into a small set of generic motifs: palatial interiors, cathedrals, villages at dusk. They read this as attractor dynamics and attributed the motifs to statistical regularities in the training data (Hintze et al., Patterns, 2025).

Their limitations section is candid about what was not measured: basin size, residence time, escape and transitions between states, and the role of each model. One 1,000-iteration run that reproduced the same image after step 100 is the only evidence on long-term behaviour. The language model explained about 14% of drift variance and the image generator 0.2%, which they took as a sign that the describer matters more.

Two neighbouring lines of work frame the question differently. In text-only transmission chains, Perez and colleagues treat the same phenomenon as cultural attractors and show that open-ended instructions produce stronger attraction than constrained ones (Perez et al., ICLR 2025). In the model-collapse literature the loss of diversity is a training-time problem. The loop studied here is an inference-time problem: nothing is retrained, and the attractors are properties of two fixed models talking to each other.

This draft takes Hintze's assertions as hypotheses and measures them. The words radius and depth are used throughout in the sense a tinkerer would mean them: how likely a wandering trajectory is to fall into a state, and how likely it is to stay.

What was run

Every run uses Hintze's 100 seed prompts and his caption instruction verbatim, 100 trajectories of 100 steps, caption temperature 0.9 where the model allows it, and his generation settings (1024 px, 25 steps, guidance 5.0). The code is a thin runner around his

released scripts. Runs went on a rented RTX Pro 6000 at 2.5 to 4.5 seconds per trajectory-step. A rerun of the baseline with a different generator seed gives the run-to-run noise floor, and every run had a written prediction before it started.

run	generator	captioner	change from baseline	trajectory-steps
A0-core	SDXL	LLaVA 7B	none (Hintze replication)	10,000
A0-core-2	SDXL	LLaVA 7B	new generator seed (noise floor)	10,000
A0-photo	SDXL	LLaVA 7B	instruction asks for a plain photographic description	10,000
A0-S	SDXL	Claude Sonnet 5.5	frontier captioner	10,000
E1-cap	SDXL	Qwen2.5-VL 7B	captioner swapped	10,000
E1-gen	Hunyuan-DiT	LLaVA 7B	generator swapped	10,000
E1-both	Hunyuan-DiT	Qwen2.5-VL 7B	both swapped	10,000
E2-QZ	Hunyuan-DiT	Qwen2.5-VL 7B	instruction in Simplified Chinese	10,000
arm A	SDXL	LLaVA 7B	four candidate renders per step, one picked at random (20 trajectories)	2,000
B-img	SDXL	LLaVA 7B	four candidates, those near already-visited states rejected (20 trajectories)	2,000

Measurement. Images are embedded with DINOv2 and captions with MiniLM; neither model is used to steer anything (the rejection arm steers with SigLIP, which is never used for grading). An attractor is a cluster found by pooling every intermediate step of the cells

being compared, with a fixed membership radius of 0.20 cosine in DINOv2 and exit at 1.25 times the radius. A visit counts as persistent when it lasts five steps or more. The tools reported here are:

- **Capture rate**: entries per step spent outside the attractor. **Endpoint share**: fraction of trajectories inside it at step 99. Together these are the radius.
- **Escape hazard**: exits per step spent inside. **Residence**: mean and Kaplan-Meier median visit length. **Stationary share**: the long-run occupancy of a coarse Markov chain over attractors plus free space. Together these are the depth.
- **Exit rate by depth**: hazard binned by distance from the centroid, as a fraction of the radius.
- **Seed memory**: the share of trajectories whose step-k image is nearer to their own step-0 image than to any other trajectory's, and the step at which that falls below half.
- **Diversity**: the Vendi score of the 100 endpoints and the knee of a k-means elbow, computed the way Hintze computed it and bootstrapped over subsamples.

The atlas names are mine, assigned by looking at contact sheets. Everything downstream is computed from the embeddings and the cluster memberships, not from the names.

Result 1: attractors are temporary stops

No attractor in the baseline pair is absorbing. Each of Hintze's named motifs is entered, left and re-entered many times over 100 steps, and the exit rate stays flat from step 20 to step 99. The loop is stationary: it is not still converging, it is touring.

Baseline pair (SDXL + LLaVA), 100 trajectories of 100 steps, attractors defined on the baseline's own centroids:

attractor	endpoints of 100	steps inside	entries	re-entries	escape hazard per step	mean residence (steps)	longest visit (steps)
gothic cathedral nave	11	574	30	19	0.033	30	43
gold-and-blue palatial salon	28	1,761	226	185	0.112	9	46
luxury canopy bedroom	8	364	70	51	0.170	6	43
Victorian house with garden	5	158	36	27	0.196	5	13

attractor	endpoints of 100	steps inside	entries	re-entries	escape hazard per step	mean residence (steps)	longest visit (steps)
free (no named attractor)	48	7,143					

Three readings of the table:

- **Even the deepest basin leaks.** The cathedral is the motif Hintze's readers remember, and it is the stickiest state we found anywhere: a 30-step mean stay and 11 trajectories still inside at the end. It still loses a trajectory every 30 steps on average, and the longest completed visit is 43 steps, under half the run. Its hold is strong, not permanent.
- **The popular basin is shallow.** The salon captures far more often (226 entries against 30) and holds 28 endpoints, but a stay lasts 9 steps and most entries are returns by trajectories that left earlier. What looks like convergence to the salon in an endpoint plot is a trajectory that keeps coming back.
- **The exit rate does not fall with time.** Pooled over the named basins, the exit rate per step is 0.13 in steps 0 to 20, then 0.10, 0.12, 0.10 and 0.11 in each later 20-step window. A loop that was settling would show a falling curve.

The rerun with a new generator seed gives the noise on these numbers: salon hazard 0.112 against 0.109, cathedral 0.033 against 0.022, bedroom 0.170 against 0.200. Hazards are reproducible to about 30% for the big basins and worse for the small ones, which bounds what any single pair can say.

Hintze's anecdote of a 1,000-step run that reproduced the same image after step 100 is not contradicted by this. The question it raises is whether residence has a heavy tail, and a 100-step horizon censors the answer. That is the first experiment I would run next: ten trajectories for 1,000 steps on three or four pairs.

Result 2: radius and depth

Radius and depth separate cleanly, and they do not go together. The salon has the largest radius and a middling depth. The cathedral has a small radius and the greatest depth. Depth is not a single number for a basin but a profile that steepens toward the rim.

Radius is how likely a wandering trajectory is to fall in. Measured as capture rate per step spent outside, the salon captures at 0.028, the bedroom at 0.007, the Victorian house at 0.004 and the cathedral at 0.003. Over a 100-step run that is the difference between a

state almost every trajectory visits and one that 30 trajectories find.

Depth is how likely a trajectory is to stay. Measured as the escape hazard, the cathedral at 0.033 per step is three times deeper than the salon at 0.112 and six times deeper than the Victorian house at 0.196. The coarse Markov chain over the named basins puts 19% of long-run occupancy in the salon and 9% in the cathedral, with 67% in free space.

The depth profile. Binning every inside-step by its distance from the centroid, as a fraction of the membership radius, gives an exit rate that rises steadily from the core to the rim:

distance from centroid, as a fraction of the radius	steps	exit rate per step
0.0 to 0.4	301	0.017
0.4 to 0.6	688	0.073
0.6 to 0.8	894	0.087
0.8 to 1.0	649	0.173
1.0 to 1.25 (hysteresis band)	325	0.200

A trajectory at the heart of a basin has a 1.7% chance of leaving on the next step. At the rim it is 20%. The practical consequence for anyone trying to push a loop out of an attractor is that the push has to carry the trajectory past the rim in one move, because a small displacement leaves it in a region from which it mostly falls back.

The free share deserves a sentence. Two thirds of all baseline steps are outside every named attractor, and 48 of 100 endpoints are free. A loop that converges to a handful of motifs still spends most of its time between them. Whether that free space is itself structured, as a family of shallow states too small to pass the persistence test, is open. The rejection experiment in a later section suggests it is.

Result 3: the landscape belongs to the model pair

Swapping one model does not perturb the baseline's attractors; it replaces the half of them that model owns. The image generator sets the image-space attractors and the captioner sets the caption-space attractors. A motif can appear in several pairs, but its depth does not travel with it.

The two-by-two. Four cells share the seeds and instruction: SDXL or Hunyuan-DiT as generator, LLaVA or Qwen2.5-VL as captioner. Pooling every intermediate step of all four cells and clustering jointly gives one atlas in image space and one in caption space, with each attractor marked present in a cell when it has three or more persistent visits there.

	image space (DINOv2)	caption space (MiniLM)
baseline attractors that survive a captioner swap	7 of 11	2 of 12
baseline attractors that survive a generator swap	4 of 11	8 of 12
attractors attributable to the generator alone	5	0
attractors attributable to the captioner alone	2	9
attractors specific to one pair	13	22

The generator also sets how attractor-dominated a loop is. The two SDXL cells spend 40% and 38% of their steps inside persistent attractors; the two Hunyuan cells spend 18% and 20%. Endpoint diversity goes the same way: Vendi 32 for the baseline, 59 for the Hunyuan-LLaVA cell.

Overlap is real but thin. The cathedral appears in three of the four cells, rendered two ways. The canopy bedroom and the dancers each appear in three. When Claude Sonnet replaces LLaVA as captioner on SDXL, it creates no new image attractors of its own; it lands in the ones Qwen had found, which LLaVA never reached. The set of reachable motifs looks like a property of the generator, and the captioner chooses which of them the loop visits.

Depth does not transfer. This was the surprise. Two predictions were written before the factorial: mine, that a given attractor′s escape hazard stays within a factor of two across pairs; Claude′s, that it stays within a factor of two only when the captioner is shared. Neither held. Of 19 attractor-pair comparisons with enough visits, the hazard ratio is within two for 29% of same-captioner pairs, 44% of same-generator pairs and 33% of pairs sharing nothing. The cathedral′s hazard is 0.024 under SDXL-LLaVA and 0.204 under Hunyuan-LLaVA, an eightfold difference with the same captioner. Gravity is a property of the pair at the attractor, not of the attractor.

Why this matters for the field's framing: the describer explaining more drift variance, as in Hintze's analysis, is true of the words and not of the pictures. Each model's share of the dynamics is visible only in its own modality, and a single-modality measurement will attribute the whole loop to whichever model it happens to look at.

Result 4: the instruction and its language reshape the landscape

Changing one sentence of the caption instruction, or its language, moves basins without changing the rate of convergence. The new basins were in most cases already latent in other cells at low visit counts. The instruction appears to select which of a generator's reachable modes get amplified.

Photographic instruction. Replacing "Describe this image in vivid, artistic terms" with "Describe this image as a realistic photograph, in plain, precise terms" on the baseline pair left plateau and knee unchanged. The salon basin halved (steps inside 1,761 to 871, hazard 0.11 to 0.22). The cathedral was untouched (574 to 580 steps, hazard 0.033 to 0.040). Three attractors appeared that no other cell has: a fitted kitchen, a supermarket produce aisle, and sepia portraits of Native Americans in headdress. Endpoint diversity rose, Vendi 32 to 44, and the share of steps inside any persistent attractor fell from 37% to 30%. Asking for photographs made the loop less attractor-bound and moved it toward stock-photography subjects.

Chinese instruction. Translating the instruction into Simplified Chinese on the Hunyuan-Qwen pair, with the same English seeds, kept convergence and register: plateau 19 against 25, Vendi 36 against 39, and 52% of persistent visits in the English cell's top five attractors. The content moved. Faces became East Asian within one to five steps. Three basins appeared that the English cell lacks: a pagoda by a misty lake (17 visits), an arched stone bridge under a full moon (12), a lone boat on a misty lake (27). Four of the English cell's attractors dropped out.

Latent modes. The pagoda lake had three visits under the English Qwen captioner on SDXL before the Chinese instruction made it a basin of 17. The Native American portraits of the photo cell sit near the baseline's vintage-photograph states. In both cases the instruction did not create a new mode; it raised the capture rate of one that the generator could already render. The testable form of this: a mode's presence at low count in one cell predicts whether an instruction change can turn it into a basin in another. We have two instances, which is a hypothesis, not a result.

A measurement instruction is itself an intervention. In a separate experiment on demographic attributes, adding one sentence that asked the captioner to state age, gender and skin tone created a brand-new attractor: grids of dozens of faces, in 55% of images by step 19, from a phrase the captioner began producing once two people were in frame. Anyone who adds an instruction to probe a loop should expect to have changed its landscape.

Result 5: what does not change

The seed is forgotten at the same rate in every cell. Across eight 100-step runs spanning two generators, three captioners, two instructions and two languages, the step at which fewer than half the trajectories are still nearer their own starting image than any other trajectory's falls between 3 and 6. The knee of the endpoint elbow is 8 to 10 everywhere, including Hintze's own published endpoints under the same rule.

run	seed half-life, images (steps)	seed half-life, captions (steps)	share still identifiable at step 20
A0-core (SDXL + LLaVA)	4	3	0.12
A0-core-2 (rerun)	6	4	0.24
A0-photo	5	5	0.21
A0-S (SDXL + Sonnet)	4	4	0.20
E1-cap (SDXL + Qwen)	3	3	0.15
E1-gen (Hunyuan + LLaVA)	3	4	0.13
E1-both (Hunyuan + Qwen)	5	4	0.12
E2-QZ (Chinese instruction)	4	not measurable in an English sentence model	0.10

Chance identification is 0.01. A fifth of trajectories remain faintly identifiable at step 20, so memory decays but does not vanish; what survives is usually a palette or a composition rather than a subject.

This is the one quantity that behaved like a constant of the loop rather than a property of the pair. The forgetting rate is set by the channel, a caption of a few dozen words passed through a generator that samples freely, and not by which models fill the roles. The attractors are where a trajectory goes after it has forgotten; the forgetting itself looks universal.

Speculation: why these attractors

Hintze's hypothesis is that the motifs are the dense regions of the training data. Nothing here tests that directly, but three observations fit it and one experiment would test it.

- **The motifs follow the generator, and the generators have different training corpora.** SDXL's basins are interiors a stock-photo library is full of: palatial salons, canopy bedrooms, luxury bathrooms, cathedral naves. Hunyuan-DiT, trained on a bilingual corpus, has its own family: dancers in swirling gowns, rainbow tunnels, a lone tree in a mirror lake. The Chinese instruction woke pagodas and moon bridges in Hunyuan that the English instruction left at trace levels. If attractors were artefacts of the loop mechanics alone, the two generators would share them.
- **The instruction steers toward a region of the corpus, and the loop amplifies what it finds there.** The photographic instruction produced kitchens and supermarket aisles, which is what plain photographic captions describe in a web-scraped corpus. The artistic instruction produced "ethereal" interiors, which is what captions with those words describe. The loop reads like a Markov chain on caption vocabulary whose stationary distribution is wherever the captioner's register and the generator's density agree.
- **The captioner contributes a register, not subjects.** LLaVA's loops are built from "vivid, ethereal, serene". Qwen writes short literal captions and reaches mirror lakes. Sonnet drifted the whole run into a purple-magenta palette, the strongest colour attractor of any cell, from nothing in the instruction. Each captioner's default vocabulary acts as a bias field on the generator's density.

The test. For each attractor centroid, estimate the density of the generator's training data near it. For SDXL the LAION corpus is public enough to retrieve nearest neighbours by CLIP similarity and count how many fall within a fixed radius. The prediction is that capture rate, and perhaps depth, rises with that density, and that attractors present

under one instruction but not another sit in regions whose captions carry that instruction's vocabulary. I have not done this and would want help from someone who has worked with corpus retrieval at that scale.

A second, cheaper test is the latent-mode prediction from the previous section: catalogue every low-count cluster in every cell, then check whether basins that appear under a new instruction or language were already present at trace level. Two of two so far.

Keeping loops out of attractors

One intervention has been tried so far, and it failed in an informative way: rejecting images that land near already-visited states makes the loop leave attractors faster and re-enter other members of the same family. This section is the start of the conversation, not its conclusion.

What was tried. On the baseline pair, every step produced four candidate captions and four renders. A control arm picked one at random. The rejection arm kept an archive of every state that had stayed put for five steps, discarded any candidate within one persistence radius of an archive entry, and picked a survivor at random. When all four were rejected it resampled up to twice more and then advanced the most distant candidate. Twenty trajectories, 100 steps, 16,000 candidate images.

What it did.

	control arm	rejection arm
share of steps inside a persistent attractor	46%	33%
median stay (steps)	10	7
salon escape hazard per step	0.20	0.36
bedroom escape hazard per step	0.14	0.33
endpoints inside a persistent attractor	55%	60%
endpoint diversity (Vendi, 20 trajectories)	13.8	12.7
steps where every candidate was rejected, last quarter		5.6%

The rejection raised every escape hazard two- to three-fold and shortened every stay. It did not change where the trajectories ended. They toured salon, cathedral, castle and back, and the archive grew by one entry every 2.6 steps as the loop kept finding fresh states next to the ones already stored. The four candidates at a step differ from each other by about 1.7 rejection radii, so the sampler was offering real choice; the limit is that the radius was sized to a five-step persistence window while the attractor family is many radii across. Discarding the points already visited leaves the trajectory inside the family.

The depth profile in Result 2 says why. A push of one radius moves a trajectory from the core to the rim, where the exit rate is 20% per step and the return rate is 80%. The push has to be larger than the basin, which the persistence-scale rule never attempts.

What to try next, in the order I would run them.

1. **Basin-scale repulsion.** Reject candidates by distance to the centroid of any known basin, with the radius taken from the basin's measured extent rather than from a persistence window. This is the direct fix for the scale mismatch. The risk is that it just reveals the next shell of shallower states; the free-space question from Result 2 gets answered either way.
2. **Caption-stage rejection.** The caption is the bottleneck through which memory passes, and the captioner owns the caption-space attractors. Rejecting captions near archived captions, before any rendering, is cheaper per step than rendering four images and attacks the modality where the loop's vocabulary collapses.
3. **Instruction design as a lever.** The photographic instruction cut the share of steps in attractors from 37% to 30% and raised diversity by a third, for the cost of one sentence. Every instruction has its own attractors, so this is a trade, but it is the cheapest trade available and it composes with the others.
4. **A human in the loop, sparsely.** In a 30,000-agent social network of AI agents, the agents that stayed novel were the ones whose humans kept supplying distinctive material and revised it when output narrowed (Lai et al., 2026). The loop version is a human who replaces the caption every k steps. The experiment is to find the largest k that keeps the trajectory out of the family, which also measures how much human attention an open-ended loop needs.
5. **Novelty rewards and temperature**, which Hintze proposed. Temperature in his sweep did not remove convergence, so I rank these last, but they are the easiest to run on the existing code.

Limitations and open questions

- **The horizon censors depth.** At 100 steps the cathedral's longest completed visit is 43 steps and 11 trajectories are still inside at the end. Whether residence has a heavy tail is unknown, and Hintze's one long run hints that it might. Ten trajectories for 1,000 steps on three or four pairs would settle it for about the cost of one 100-step cell.
- **Hazards carry about 30% noise for the big basins and more for small ones.** The rerun with a new generator seed is the only replicate. Ratios under two between cells should be read as ties.
- **The membership radius is a choice.** Every atlas uses 0.20 cosine in DINOv2 with exit at 1.25 times that, calibrated once on the baseline. Attractor counts and shares move with it; hazard orderings have been stable across the values tried, but that is not a sensitivity analysis.
- **Two generators, three captioners, one architecture family each.** The generator-owns-images result rests on one swap in each direction. A third generator would say whether it generalises.
- **The caption-space tools are English.** The Chinese cell's caption-space numbers are uninterpretable and are not used; its image-space numbers are fine.
- **Attractor names are mine**, assigned from contact sheets. Nothing numeric depends on them, but a reader who disagrees with a name should check the sheet.
- **Claude did the engineering and the analysis.** I set the questions, wrote or confirmed every prediction before each run, labelled a check set by hand, and read every report; Claude Code wrote the runner, the analysis scripts and the first drafts of the reports and of this document. The pre-registration guards against fitting the story to the data, but it does not guard against both of us sharing a blind spot. That is one reason this draft is being circulated.
- **The free space is unexplained.** Two thirds of baseline steps are outside every named attractor. The rejection experiment suggests the free space is a dense shell of shallow states, but the atlas rules were built to find persistent basins and would miss that structure by design.

Open questions I would most like help with: whether the depth profile is universal across pairs; whether training-data density predicts capture rate; what the right persistence rule is for shallow states; and whether a basin-scale push produces exploration or a new family of attractors.

Acknowledgement

This project was a collaboration between the author and Claude (Anthropic). Claude Fable 5.1 was used through Claude Code as engineer, analyst and writing partner, and Claude Sonnet 5.5 through the API as one of the captioners under test. The author set the questions, chose and sized the experiments, wrote or confirmed every prediction before each run, named the attractors from the contact sheets and read every report. Claude wrote the runner and the analysis code, drafted the quantitative thresholds of the predictions, proposed several of the measurements (the persistent atlas, the escape-hazard and capture-rate readouts that implement the author's gravity idea, the replication noise floor), built the reports and wrote the first draft of this document on 2026-10-23, which the author reviews and revises; the line under the title says where that stands. Where Claude's quantified predictions and the author's directional ones differed, both are recorded in the repository with the outcomes. Every number here was computed by code in the repository from the run data. The author takes responsibility for the claims. A separate note describes the collaboration in detail.

Data, code and references

Code and data. All runs, configs, predictions, analysis scripts and gate reports are in a git repository (github.com/bradhinkel/ai_to_ai_loops, private while this draft circulates; access on request, to be opened with the paper). Every image and caption of every run is kept, with the model versions, Ollama digests, generator seeds and package versions logged in each step record. The runner is a thin wrapper around Hintze's released scripts; every departure from his settings is listed in a deviations file with its reason.

Models. Stable Diffusion XL base 1.0; LLaVA 7B (Ollama llava:latest, Q4_0); Qwen2.5-VL 7B (Ollama qwen2.5vl:7b); Hunyuan-DiT v1.2; Claude Sonnet 5.5 via the Messages API. Evaluation embeddings: DINOv2 for images, all-MiniLM for captions. Steering in the rejection arm: SigLIP, never used for evaluation.

References.

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Companion draft. Bias in AI-to-AI loops: what happens to seeded demographic attributes, and whose defaults replace them.